

RFID BASED ATTENDANCE SYSTEM

MINI PROJECT (ECD334) REPORT

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to

APJ Abdul Kalam Technological University

In partial fulfilment of the requirements for the award of the

Degree of

Bachelor of Technology

in

ELECTRONICS AND COMMUNICATION ENGINEERING



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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March 2026

DECLARATION

*We hereby declare that the project report "**RFID BASED ATTENDANCE SYSTEM**", submitted for partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala** is a bonafide work done by us under the supervision of **Dr. Jinsa Kuruvilla and Associate Prof. Dr. Siddharth Shelly**. This submission represents our ideas in our own words and where ideas or words of others have been included. We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.*

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CERTIFICATE

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ABSTRACT

The RFID-based attendance system project report presents a comprehensive and efficient solution for tracking attendance in various settings. This advanced system leverages Radio Frequency Identification (RFID) technology to automate the attendance process, ensuring accuracy and reliability. By utilizing RFID tags or cards, the system is capable of uniquely identifying individuals, thereby eliminating the need for manual attendance recording and significantly reducing the chances of errors.

Traditional attendance tracking methods often prove to be time-consuming and inefficient. The Internet of Things (IoT) serves as a crucial medium in addressing this issue, facilitating the automation of processes. Among the various automatic identification technologies, RFID has gained widespread popularity due to its effectiveness. RFID is a wireless communication technology designed for the purpose of identifying and tracking objects using radio waves. It operates by transferring data from an electronic tag, commonly referred to as an RFID tag or label, to an RFID reader, thereby enabling seamless data collection and processing.

This IoT-based project is specifically developed to reduce the reliance on manual efforts, enhance automation, and offer significant benefits to educational institutions such as schools and colleges, as well as workplaces and other professional environments. By integrating RFID technology, this system not only simplifies attendance tracking but also enhances efficiency and reliability in managing attendance records.

In summary, the RFID-based attendance system project report provides in-depth insights into the design, development, implementation, and assessment of an advanced attendance management solution. It highlights the practical applications of RFID technology in streamlining attendance tracking and offers valuable implications for organizations aiming to optimize their attendance management processes.

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Chapter 1

INTRODUCTION

In today's digital era, traditional attendance tracking methods like manual sign-in sheets and biometric scanners are becoming inefficient and error-prone. These methods often lead to inaccuracies, consume time, and require manual effort, making them less practical for large-scale use. To overcome these limitations, an automated attendance system using Radio Frequency Identification (RFID) technology offers a highly effective solution.

RFID technology uses electromagnetic fields to automatically identify and track tags attached to objects or individuals, enabling a seamless and contactless attendance monitoring system. By eliminating manual record-keeping and reducing human intervention, this approach enhances accuracy, efficiency, and convenience, making it ideal for educational institutions, workplaces, and other professional settings.

This mini-project focuses on implementing an RFID-based attendance system to provide a streamlined, automated, and technologically advanced solution for managing attendance records with greater precision and ease.

1.1 SCOPE

The scope of this mini-project extends beyond basic attendance automation, aiming to develop a comprehensive RFID-based system that caters to the specific needs of educational institutions, corporate workplaces, and other organizations looking to modernize their attendance management process. This project encompasses the design and implementation of both hardware and software components essential for accurate and efficient attendance tracking. Additionally, it considers key aspects such as scalability, ensuring the system can accommodate a growing number of users, and compatibility with existing infrastructure to facilitate seamless integration. Another critical focus is the development of a user-friendly interface that enhances accessibility and ease of use for both administrators and end-users. By addressing these factors, this project aims to provide a practical, efficient, and technologically advanced solution for attendance management.

1.2 OBJECTIVE

This mini-project aims to modernize the attendance management process by implementing an automated RFID-based system. The transition from traditional manual methods to a

technology-driven approach is intended to enhance accuracy, efficiency, and security while providing comprehensive data insights. The primary objectives of this system are as follows:

1. Elimination of Human Error

Manual attendance tracking is susceptible to inaccuracies such as incorrect entries, duplication, and data loss. By automating the process, this system ensures precise and error-free attendance records, thereby minimizing discrepancies and reducing the need for manual corrections.

2. Improved Time Efficiency

Conventional attendance recording methods, including roll calls and sign-in sheets, require significant time and effort. This RFID-based system optimizes the process by allowing attendees to scan their RFID tags, thereby significantly reducing the time required for attendance marking and improving overall operational efficiency.

3. Enhanced Security and Authentication

The integration of RFID technology ensures that attendance data remains secure through encryption and authentication mechanisms. This prevents unauthorized access, restricts attendance marking to registered individuals, and safeguards sensitive information from tampering or misuse.

4. User Convenience and Accessibility

The system offers a streamlined and user-friendly experience by eliminating the need for manual sign-ins. Attendees can efficiently mark their attendance by scanning their RFID tags, making the process more convenient and reducing administrative burdens.

1.3 SYSTEM DESIGN

The proposed design for the automated attendance system incorporates multiple interconnected components, each serving a vital function to ensure the system's overall efficiency, accuracy, and reliability. The key components of this system are as follows:

1. RFID Tags

Each individual is assigned a unique RFID tag. These tags may be in the form of cards or keychains and contain a microchip and antenna. The microchip stores a unique identification number (UID), while the antenna enables wireless communication with the RFID reader.

2. MFRC522 RFID Reader Module

The MFRC522 module is used to read the RFID tags. It emits radio frequency signals and detects nearby RFID cards. When a card is brought close, the module reads its UID and sends this data to the microcontroller for processing.

3. ATmega328P Microcontroller

The ATmega328P acts as the brain of the system. It receives UID data from the RFID reader and compares it with the predefined list of authorized IDs stored in the program (Arduino IDE). Based on this comparison, it determines whether access is granted or denied.

4. LCD Display

The LCD display provides visual feedback to the user. It displays messages such as “Access Granted”, “Access Denied”, or “Already Marked”, making the system user-friendly and easy to interpret.

5. Buzzer

The buzzer provides audio indication for when valid attendance is marked.

By integrating these components into a cohesive system, the proposed RFID-based attendance system enables a simple and efficient method for attendance marking. When a user scans their RFID card, the system checks whether the UID is authorized. If valid, it displays “Access Granted” and triggers the buzzer. If the card is not recognized, it displays “Access Denied”. If the same authorized card is scanned again, the system indicates “Already Marked”.

This approach eliminates manual attendance marking, reduces errors, and ensures quick and reliable operation without the need for network connectivity or database storage.

Although the current system operates as a standalone embedded solution, it offers strong scalability. Future enhancements can include integration with databases, cloud connectivity, biometric authentication, or mobile applications for real-time monitoring and advanced analytics. This flexibility makes the system adaptable for use in educational institutions, offices, and other controlled access environments.

Chapter 2

LITERATURE SURVEY

1. Research Paper Title: “RFID Based Attendance System”

- **Authors:** Aman Saxena

- **Published in:** International Journal for Modern Trends in Science and Technology, January 2021

Summary: This paper discusses the design and implementation of an RFID-based attendance system aimed at enhancing the efficiency and accuracy of attendance tracking. The system utilizes RFID cards and readers to automate the identification process, thereby reducing the time and errors associated with manual attendance methods. The author highlights the system’s potential to streamline attendance management in educational institutions and corporate environments.

2. Research Paper Title: “RFID Based (IoT) Automatic Attendance System: A Survey Analysis”

- **Authors:** R. K. A. R. Kariapper and M. S. Suhail Razeeth

- **Published in:** Southeastern University International Conference on Computing and Engineering, 2019

Summary: This study analyzes the development and implementation of an Internet of Things (IoT) based attendance system using RFID technology. The authors emphasize the advantages of RFID in automating attendance processes, noting its higher accuracy and performance compared to traditional paper-based systems. The paper also discusses the integration of RFID with IoT to enable real-time attendance tracking and data management.

3. Research Paper Title: “Attendance and Information System using RFID and Web-Based Technologies”

- **Authors:** Kashif Ishaq and Samra Bibi

- **Published in:** International Journal of Advanced Computer Science and Applications (IJACSA), 2018

Summary: This paper proposes an RFID-based Attendance Management System (AMS) integrated with web-based technologies for academic environments. The system comprises RFID readers, programmable logic circuits such as Arduino, and a web application to facilitate real-

time attendance tracking and information dissemination. The authors discuss the system's architecture, implementation challenges, and potential benefits in enhancing attendance management and communication within educational institutions.

4. Research Paper Title: "An RFID-Based Attendance System with Web-Based Monitoring"

- **Authors:** M. M. Rahman, M. I. Hossain, and S. K. Alam

- **Published in:** International Journal of Computer Applications, 2012

Summary: This paper presents a novel RFID-based attendance system integrated with a web-based monitoring platform. The system architecture includes RFID readers, tags, a MySQL database, and a web application for real-time attendance tracking and reporting. The authors discuss the design principles, implementation challenges, and usability evaluation of the system, highlighting its potential applications in various sectors, including education and corporate environments.

5. Research Paper Title: "RFID Based Attendance System for Department"

- **Authors:** Krupali Panchal, Parthvi Jingar, Om Vataliya, and Harshul Rathod

- **Published in:** International Journal of Innovative Science and Applied Engineering, 2022

Summary: This study presents a cost-effective RFID-based attendance tracking system designed for departmental use within educational institutions. The authors focus on minimizing implementation costs by utilizing low-power RFID readers and open-source database solutions. The system is evaluated based on its reliability, affordability, and ease of deployment. The findings suggest that even small departments can adopt RFID technology for attendance management without significant financial investment.

Chapter 3

METHODOLOGY

This project consists of two phases:

1. **Phase 1:** Familiarization with Altium Designer and PCB fabrication by designing and fabricating a RC phase shift oscillator using an op-amp with a clipper.
2. **Phase 2:** Development of RFID Based Attendance System using Arduino Uno.

Each phase follows a structured approach, including circuit design, PCB fabrication, component integration, programming, and testing.

Phase 1: Familiarization with PCB Fabrication (RC phase shift oscillator using an op-amp with a clipper Circuit using Op-Amp)

1. Understanding PCB Fabrication Process

- Researching and learning PCB design principles, fabrication steps, and materials.
- Understanding Altium Designer software for schematic capture and PCB layout.

2. Circuit Design and Simulation

- Designing a RC phase shift oscillator using an op-amp (LM741) with a clipper circuit with appropriate resistors and capacitors.
- Simulating the circuit in Altium Designer to verify expected behaviour.

3. PCB Layout Design

- Creating the PCB layout in Altium Designer by placing components and routing connections.
- Generating Gerber files for PCB fabrication.

4. PCB Fabrication and Assembly

- Printing the PCB layout and etching the copper-clad board using Ferric Chloride solution.
- Drilling holes for through-hole components.
- Soldering the components onto the fabricated PCB.

5. Testing and Verification

- Testing the circuit and observing the output on digital storage oscilloscope.

- Verifying whether the circuit generates the clipped sine waveform at designed frequency.

Phase 2: RFID Based Attendance System using Arduino Uno

1. System Design and Planning

- Defining the system architecture, including RFID detection, UID identification and attendance recording.
- Selecting suitable components based on project requirements.

2. Component Selection

- ATmega328P Microcontroller
- 16 MHz Crystal Oscillator
- Alphanumeric LCD (16×4 with I2C)
- RFID Reader MFRC522 Module
- RFID Tags / RFID Cards

3. Fabrication and Assembly

- Soldering components and ensuring proper connections.

4. Programming and Software Development

Writing Arduino code in C++ (Arduino IDE) for:

- Identifying unique ID of each card.
- Reading data from MFRC522 RFID reader.
- Triggering buzzer.
- Sending data serially to an excel sheet.

5. Testing and Performance evaluation

- Testing the proper detection of each card.
- Ensure count and access approval will be displayed on lcd.

3.1 BLOCK DIAGRAM:

RFID Based Attendance System:

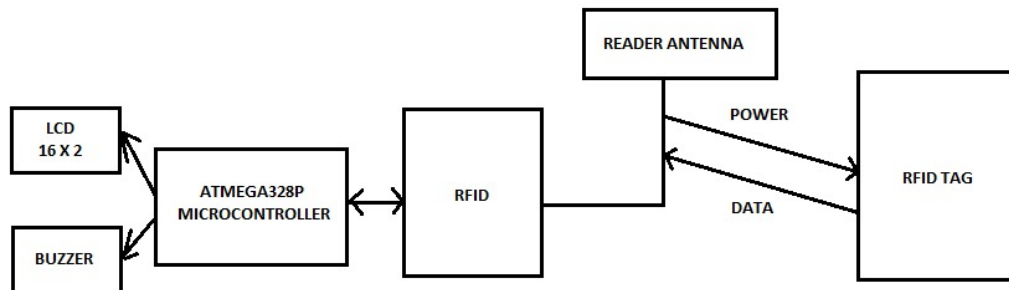


Figure 3.1: Block diagram of the whole system

The block diagram represents the working of the RFID-based Attendance System using AT-MEGA32P. In this system, the ATMEGA32P functions as the central processing unit, coordinating and managing the operation of all other components, which act as input and output units. The roles of the different blocks in the diagram are explained as follows:

- **RFID Reader:** This serves as the input unit of the system. It is responsible for detecting and reading data from an RFID tag or card. When a user presents an RFID card, the reader captures the unique identification data encoded within it and transmits this information to the microcontroller for further processing.
- **Microcontroller Block:** The ATMEGA328P, acting as the microcontroller, serves as the core data processor and central controller of the entire system. It receives the tag data from the RFID reader, processes it to verify the identity of the cardholder and sends details serially to the external system.
- **LCD display and buzzer:** Once the microcontroller successfully verifies the received RFID tag data, the attendance will be granted and same will be displayed on LCD screen, buzzer also sounds when access granted.

Chapter 4

SYSTEM OVERVIEW

4.1 PHASE 1

TRAPEZOIDAL WAVE GENERATOR USING RC PHASE SHIFT OSCILLATOR (OP-AMP)

4.1.1 COMPONENTS

- Operational Amplifier(Lm741)
- Resistors
- 1 M Ω Potentiometer
- Capacitors
- 1N4001 Diodes
- Dual DC Power Supply
- DC Reference Supply
- Connecting Wires and Headers

4.1.2 COMPONENTS DESCRIPTION

RC PHASE SHIFT OSCILLATOR

741 OP AMP

The $\mu A741$ is a general-purpose operational amplifier widely used in analog electronic circuits for amplification, filtering, signal conditioning, and oscillator applications. It is a high-gain, direct-coupled amplifier that operates using differential input signals and produces a single-ended output. The 741 op-amp consists of multiple transistor stages internally, including a differential amplifier stage, an intermediate gain stage, and an output stage. It is designed to amplify the difference between the voltages applied at its inverting and non-inverting input terminals while rejecting any common-mode signals. The amplifier requires dual power supply voltages for proper operation and provides a high open-loop gain, high input impedance, and low output impedance. In oscillator circuits such as the RC phase shift oscillator, the 741 op-amp is used as an inverting amplifier, providing a phase shift of 180° . When combined with an RC feedback network that introduces an additional 180° phase shift, the total phase shift becomes 360° , satisfying the phase condition for sustained oscillations. Due to its stable

performance and ease of use, the $\mu A741$ is commonly employed in laboratory experiments and educational applications.

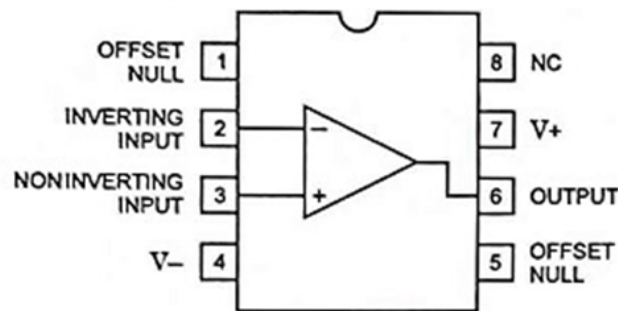


Figure 4.1.1: Pin diagram of 741 OP AMP

KEY FEATURES

- High open-loop voltage gain
- High input impedance and low output impedance
- Differential input and single-ended output
- Internally frequency compensated for stability
- Operates with dual power supply ($\pm V_{CC}$)
- Low offset voltage and low bias current
- Wide range of applications including amplifiers, oscillators, and filters
- Short-circuit protection at the output
- Low noise and reliable performance
- Available in 8-pin DIP package

POTENTIOMETER (1 M Ω)

A potentiometer of value 1 M Ω is used in the RC phase shift oscillator circuit to provide adjustable resistance. It is a three-terminal variable resistor that allows fine control of the circuit parameters. In this circuit, the potentiometer is mainly used to adjust the effective gain of the amplifier. By varying the resistance, the loop gain of the oscillator can be slightly increased or decreased, which helps in initiating oscillations and stabilizing the output amplitude. The use of a high-value potentiometer ensures smooth adjustment without significantly loading the feedback network. This helps achieve sustained oscillations with minimal distortion.



Figure 4.1.2: 1 MΩ Potentiometer

RESISTORS

Resistors are used in the circuit to set the gain of the amplifier and to form the RC phase shift network. A resistor of value $3.3\text{ k}\Omega$ is used in each section of the RC network. These resistors, along with the capacitors, determine the frequency of oscillation and contribute to the required phase shift. A $33\text{ k}\Omega$ resistor is used as an offset or biasing resistor to ensure proper operating conditions for the operational amplifier and to maintain stability. An $18\text{ k}\Omega$ resistor is placed in the input path to the amplifier after the feedback circuit to control the input signal level and to help set the overall gain of the amplifier. Proper selection of resistor values is essential to satisfy the Barkhausen criterion for sustained oscillations.



Figure 4.1.3: Resistor

CAPACITORS

Capacitors are used in conjunction with resistors to form the RC phase shift network of the oscillator. Each capacitor works with its corresponding resistor to introduce a phase shift that varies with frequency. The three cascaded RC sections together provide a total phase shift of 180° at the desired frequency of oscillation. The capacitors also help in frequency selection by

allowing only a specific frequency to pass through the feedback network with the correct phase relationship. The values of the capacitors are chosen such that stable sinusoidal oscillations are obtained in the required frequency range. Accurate capacitor selection ensures predictable oscillation frequency and smooth waveform generation.



Figure 4.1.4: Capacitor

TRAPEZOIDAL CLIPPER

DIODES AND REFERENCE VOLTAGE SOURCE

The diodes in the trapezoidal clipper are used to limit the positive and negative peaks of the sinusoidal waveform obtained from the RC phase shift oscillator. When the input signal exceeds the preset reference voltage level, the corresponding diode becomes forward biased and conducts, preventing any further increase in output voltage. The reference voltage source provides the threshold levels at which clipping occurs, thereby defining the flat portions of the trapezoidal waveform. Together, the diodes and reference voltage enable controlled amplitude limiting and waveform shaping.

RESISTOR

The resistor connected in series with the input of the trapezoidal clipper limits the current flowing through the diodes during conduction. It also controls the slope of the rising and falling edges of the waveform, contributing to the trapezoidal shape of the output. Proper selection of this resistor ensures safe diode operation and smooth clipping action without excessive distortion.

4.2 PHASE 2

RFID BASED ATTENDANCE SYSTEM using ATmega328P

4.2.1 COMPONENTS

- ATmega328P Microcontroller
- 16 MHz Crystal Oscillator
- Alphanumeric LCD (16×4 with I2C)
- RFID Reader MFRC522 Module
- RFID Tags / RFID Cards
- Capacitors
- Resistors
- Breadboard and Jumper Wires

4.2.2 COMPONENTS DESCRIPTION

ATMEGA328P MICROCONTROLLER

The ATmega328P is the high-performance, 8-bit AVR RISC-based brain found at the heart of the Arduino UNO. It features 32 KB of ISP flash memory, 1024 B of EEPROM, and 2 KB of SRAM for data storage. This chip can execute powerful instructions in a single clock cycle, achieving throughputs of up to 20 MIPS. In a standalone setup, this single chip replaces the entire Arduino board to save space and reduce costs. It operates between 1.8 V and 5.5 V, making it ideal for battery-powered applications. The chip also includes 23 general-purpose I/O lines and an integrated 10-bit analog-to-digital converter.



Figure 4.2.1: ATmega328P Microcontroller

16 MHZ CRYSTAL OSCILLATOR

The 16 MHz crystal oscillator is a critical component that acts as the heartbeat or clock for the microcontroller. It generates a stable and precise electronic signal that dictates the speed at which the ATmega328P executes instructions. Without this external clock source, the microcontroller would rely on its less accurate internal RC oscillator. It is typically connected to the chip with two small 22 pF ceramic capacitors to ensure stable oscillation. This specific frequency, 16 MHz, is the standard for Arduino-compatible projects, ensuring timing-sensitive functions like serial communication work correctly. Its high precision is vital for applications requiring exact timekeeping or high-speed data transfer.



Figure 4.2.1: 16 MHz Crystal Oscillator

ALPHANUMERIC LCD (16×4 WITH I2C)

A 16×4 alphanumeric LCD is a liquid-crystal display that can show up to 64 characters across four lines, with 16 characters per line. It provides a clear visual interface for displaying system status, such as Access Granted or a user's name in an attendance system. The I2C version includes a small backpack module that reduces the required wiring from 16 pins down to just 4 pins. This saves valuable I/O ports on the microcontroller for other sensors or components. The display typically features a backlight, often green or blue, for visibility in low-light conditions. The screen contrast can also be adjusted using a built-in potentiometer on the I2C module.



Figure 4.2.2: Alphanumeric LCD

ARDUINO UNO

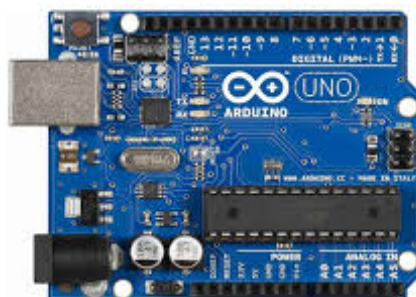
In this project, the Arduino UNO is not used as the primary controller during operation but serves as a supporting tool for programming and testing the standalone ATmega328P microcontroller.

The Arduino UNO is based on the ATmega328P microcontroller and provides an easy platform for code development and uploading using the Arduino IDE. It operates at 5V with a 16 MHz clock and includes built-in USB-to-serial communication, making it highly convenient for programming purposes.

In our system, the Arduino UNO is used in the following ways:

- **Bootloader Burning:** The Arduino UNO is configured as an ISP (In-System Programmer) to burn the bootloader onto the standalone ATmega328P microcontroller.
- **Code Uploading:** It is used to upload the program directly into the ATmega328P using SPI communication (MOSI, MISO, SCK, RESET pins).
- **Power Supply:** The UNO provides regulated 5V and 3.3V supply to power the custom-designed PCB during testing and development.
- **Development and Debugging:** It acts as a convenient interface for testing circuits before transferring the design to the final PCB.

Thus, the Arduino UNO plays a dual role in this project, functioning both as a development platform and as a part of the final implementation. It is used for bootloader burning and code uploading to the standalone ATmega328P, while also providing a stable 5V and 3.3V power supply to the custom PCB. By combining programming support with reliable power delivery, the Arduino UNO acts as a bridge between development and deployment, ensuring smooth operation of the system. Its inclusion enhances the practicality and flexibility of the design, making the overall implementation more robust and convenient.



*Figure 4.2.1: Arduino UNO Board***RFID READER MFRC522 MODULE –**

RFID (Radio Frequency Identification) is a wireless communication technology that operates at 13.56 MHz and is commonly used for identification and authentication purposes. It can communicate with microcontrollers using SPI, I²C, or UART interfaces, making it versatile for various applications. The module typically operates at 3.3V but features 5V-tolerant inputs, allowing it to be used with a wide range of devices. One widely used RFID module, the MFRC522, offers a reading range of up to 5 cm, making it suitable for contactless applications such as access control, attendance systems, and inventory tracking. Unlike optical barcode scanners, RFID technology does not require a line of sight, meaning the RFID reader can detect tags even if they are behind objects or inside enclosures. This feature enhances its usability in environments where visibility is limited or direct access to the tag is not feasible.

*Figure 4.2.2: RFID MFRC522 Module***RFID TAG –**

RFID tags are small electronic devices used for wireless identification and tracking, commonly found in many applications. They consist of a microchip for data storage and an antenna for communication with an RFID reader. Operating at different frequencies, including low frequency (LF), high frequency (HF), and ultra-high frequency (UHF), RFID tags are designed for various use cases, from short-range authentication to long-distance tracking. These tags can be passive, active, or semi-passive—passive tags rely on the reader's signal for power, while active and semi-passive tags contain an internal battery, allowing for longer-range communication and enhanced data processing capabilities. Unlike barcodes, RFID tags store more data, support simultaneous multiple tag reading, and function without requiring direct line-of-sight, making them highly efficient for automated identification and real-time tracking in diverse environments.



Figure 4.2.3: RFID Tag

BUZZER –

A buzzer is an audio signalling device that generates sound through mechanical, electromechanical, or piezoelectric means. It operates on DC voltage and is commonly used in electronic circuits for alerts, alarms, and notifications. Buzzers can be active (producing sound when powered) or passive (requiring an external signal to generate sound). Piezoelectric buzzers, which use a vibrating piezo crystal, are widely used due to their low power consumption and compact size.



Figure 4.2.4: Buzzer

OTHER REQUIRED COMPONENTS

- **Capacitors:** Used for decoupling, filtering noise, and stabilizing the crystal oscillator.
- **Resistors:** 330 Ω resistors are typically used for LEDs, while 10 k Ω resistors act as pull-ups for the reset pin.
- **Breadboard and Jumper Wires:** Essential for prototyping the circuit without permanent soldering.

4.3 SOFTWARE IMPLEMENTATION

4.3.1 Altium Designer

Altium Designer is a comprehensive PCB design software that provides a unified platform for schematic capture, PCB layout, routing, simulation, and manufacturing preparation. It features an intuitive user interface that streamlines the design process, making it suitable for both beginners and professionals. The software supports multi-layer PCB designs, with advanced routing tools such as differential pair routing, interactive routing, and autorouting, ensuring precise and efficient trace layouts. Its real-time 3D visualization allows designers to inspect board layouts and verify component placements before fabrication.

Altium Designer also includes powerful design rule checking (DRC) and signal integrity analysis, helping to minimize errors and optimize PCB performance. It offers a vast component library with real-time supplier links and custom library management, simplifying component selection. The software supports SPICE-based circuit simulation, allowing engineers to test circuit behavior before physical prototyping.

For manufacturing, Altium Designer generates Gerber files, Bill of Materials (BOM), and assembly drawings, ensuring a smooth transition to production. It also integrates with mechanical CAD software like SOLIDWORKS, facilitating enclosure design and mechanical fit verification. Widely used in embedded systems, IoT, industrial automation, automotive, and consumer electronics, Altium Designer remains a powerful, industry-leading solution for professional PCB design and development.

4.3.2 Arduino IDE

The Arduino Integrated Development Environment (IDE) is a free, open-source software used for writing, compiling, and uploading code to Arduino microcontrollers. It features a simple and intuitive interface, making it accessible for both beginners and experienced developers. Programs, known as sketches, are written in C/C++ and follow a basic structure with `setup()` and `loop()` functions. The IDE provides built-in tools for compiling code, uploading it to the board, and debugging through the Serial Monitor. It supports a wide range of Arduino boards, including Uno, Mega, Nano, and ESP8266, and allows users to manage libraries for additional functionalities like sensor interfacing, wireless communication, and motor control. The Serial Monitor and Serial Plotter help in real-time debugging and data visualization. Additionally, the Library Manager simplifies the process of adding new libraries, enhancing project capabilities. Arduino IDE is cross-platform, running on Windows, macOS, and Linux, and also offers a web-based version for online coding. While primarily designed for ease of use, it can also

TRAPEZOIDAL WAVE GENERATOR USING RC PHASE SHIFT OSCILLATOR:

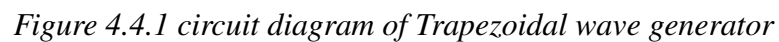


Figure 4.4.2 circuit diagram of RFID based attendance system

4.5 WORKING

4.5.1 TRAPEZOIDAL WAVE GENERATOR USING RC PHASE SHIFT OSCILLATOR (OP-AMP):

An RC phase shift oscillator is a sinusoidal oscillator that employs a network of resistors (R) and capacitors (C) to produce the required phase shift for sustained oscillations. The RC feedback network provides a total phase shift of 180° , while the amplifying device—an operational amplifier configured in the inverting mode—contributes an additional 180° . This results in a total phase shift of 360° , thereby satisfying the phase condition for oscillation as per the Barkhausen criterion. In this project, the sinusoidal output generated by the RC phase shift oscillator is applied as the input to a trapezoidal clipper circuit. The trapezoidal clipper operates by limiting the amplitude of the input signal beyond predetermined voltage levels using diodes and reference voltages. When the input sinusoidal waveform exceeds these threshold levels, the diodes conduct and clip the peak portions of the waveform, while the central portion of the waveform remains unaffected. As a result, the output waveform takes the shape of a trapezoid. Thus, the trapezoidal clipper demonstrates controlled waveform shaping by introducing non-linear behavior, allowing the conversion of a sinusoidal signal into a trapezoidal waveform for signal conditioning applications.

1. RC Phase Shift Network: The oscillator consists of three or more RC stages connected in series. Each RC stage provides a phase shift of 60° , and for oscillation, a total phase shift of 180° is required. The additional 180° phase shift is provided by an inverting amplifier (like an op-amp or transistor), resulting in a total 360° phase shift (0° net phase shift), ensuring sustained oscillations.

2. Amplification & Feedback: The amplifier (op-amp or transistor) provides gain to compensate for energy losses in the circuit. The output of the amplifier is fed back into the input via the RC network, maintaining oscillations.

3. Oscillation Frequency: The frequency of oscillation is given by:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

A clipper circuit is used at the output of the RC Phase Shift Oscillator to limit the amplitude of the generated sine wave, preventing distortion or excessive voltage levels. A diode clipper is placed at the oscillator output parallel with the load. It limits the voltage to a specific threshold (set by diodes or Zener diodes), ensuring the output remains within a desired range. Thus, combining an RC phase shift oscillator with a clipper ensures a stable and controlled sine wave output for further signal processing or transmission.

4.5.2 RFID BASED ATTENDANCE SYSTEM USING ATMEGA328P

An RFID system is made up of a reader module consisting of a transceiver and antenna, and an RFID card or tag acting as a transponder. The reader continuously generates a radio-frequency electromagnetic field through its antenna, creating a small invisible energy zone. When an RFID card enters this field, its internal antenna coil picks up the RF energy, inducing a voltage that powers the tiny microchip inside. In passive tags, this induced power acts like a battery source. Once powered, the chip accesses its stored data, such as a unique identification number.

When the transponder tag draws power in this way, the interaction of the RF fields causes the voltage at the transceiver antenna to drop in value. This effect is utilized by the tag to communicate its information back to the reader. The tag is able to control the amount of power drawn from the field and, by doing so, it modulates the voltage sensed at the transceiver according to the bit pattern it wishes to transmit.

The RFID-based attendance system uses a standalone ATmega328P microcontroller as the central controller, interfaced with an MFRC522 RFID reader, LCD display, and passive buzzer. The ATmega328P is configured in standalone mode with an external crystal oscillator, power supply, and reset circuitry, allowing it to function independently without an Arduino Uno board.

When a user places an RFID card near the MFRC522 reader, the reader detects the card and sends its unique ID to the ATmega328P through SPI communication. The microcontroller verifies the ID with the programmed code and marks attendance accordingly. Each entry is recorded with an accurate timestamp.

The LCD display shows messages such as “Access granted” when an authorized ID tries to gain access and “Access denied” when an unauthorized ID is detected. When a card corresponding to an attendance entry that has already been recorded is shown again, the LCD displays “Already marked”. It also displays the count of granted attendance. The passive buzzer sounds when access is granted.

Chapter 5

IMPLEMENTATION AND RESULT

TRAPEZOIDAL WAVE GENERATOR USING RC PHASE SHIFT OSCILLATOR:

In this project, the sinusoidal output generated by the RC phase shift oscillator is applied as the input to a trapezoidal clipper circuit. The trapezoidal clipper operates by limiting the amplitude of the input signal beyond predetermined voltage levels using diodes and reference voltages. When the input sinusoidal waveform exceeds these threshold levels, the diodes conduct and clip the peak portions of the waveform, while the central portion of the waveform remains unaffected. As a result, the output waveform takes the shape of a trapezoid. Thus, the trapezoidal clipper demonstrates controlled waveform shaping by introducing nonlinear behavior, allowing the conversion of a sinusoidal signal into a trapezoidal waveform for signal conditioning applications.

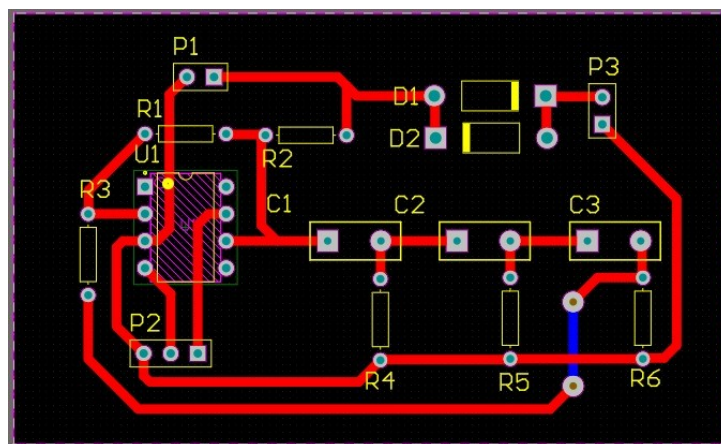


Figure 5.1 Trapezoidal Wave Generator Using RC Phase Shift Oscillator

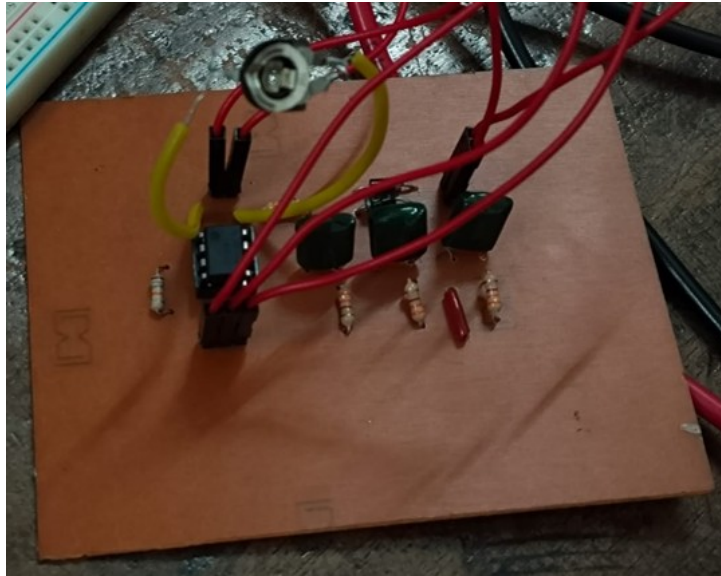


Figure 5.2 Implementation of TRAPEZOIDAL WAVE GENERATOR-USING RC-PHASE SHIFT OSCILLATOR

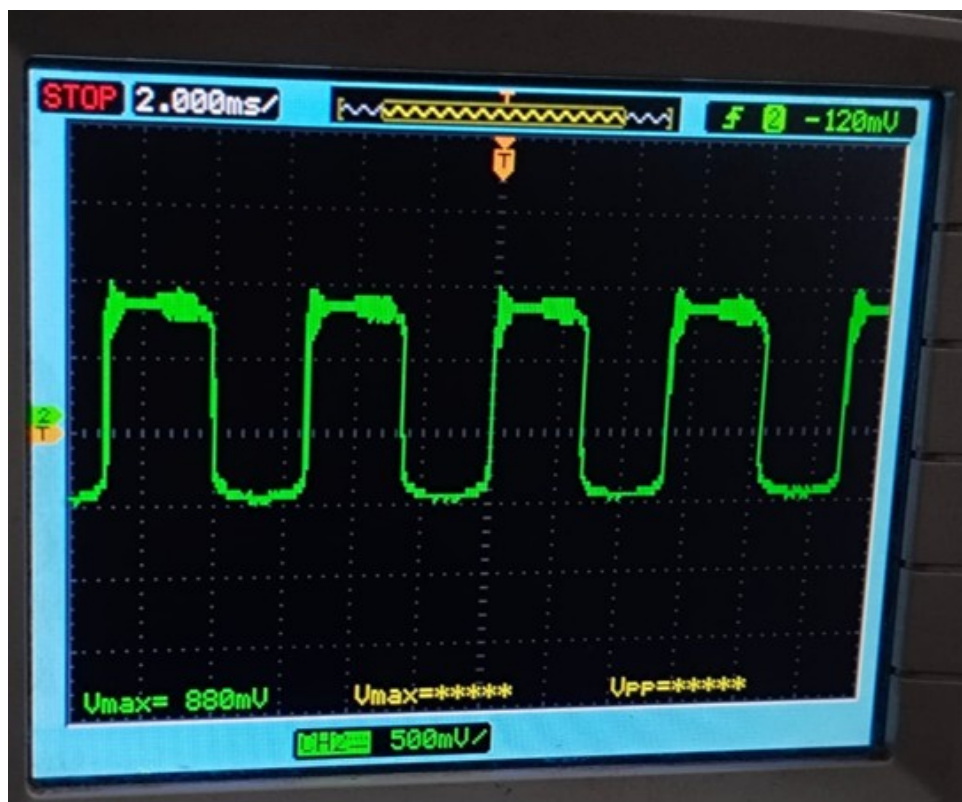


Figure 5.3 output of TRAPEZOIDAL WAVE GENERATOR-USING RC-PHASE SHIFT OSCILLATOR

RFID Based Attendance System :

The RFID-based attendance system was implemented using a custom-designed PCB, ensuring a compact and efficient hardware layout. Initially, the complete circuit was tested and verified on a breadboard to ensure correct functionality. Following this, the schematic was designed using Altium Designer, and the PCB layout was prepared. The physical PCB was fabricated using a copper-clad sheet through lamination and etching processes, followed by drilling and soldering of components such as the ATmega328P microcontroller, MFRC522 RFID module, LCD display, and buzzer. After assembly, the system was tested for proper operation. The RFID reader successfully detected tags and the system responded accurately by displaying appropriate messages, resulting in a reliable and user-friendly attendance system.

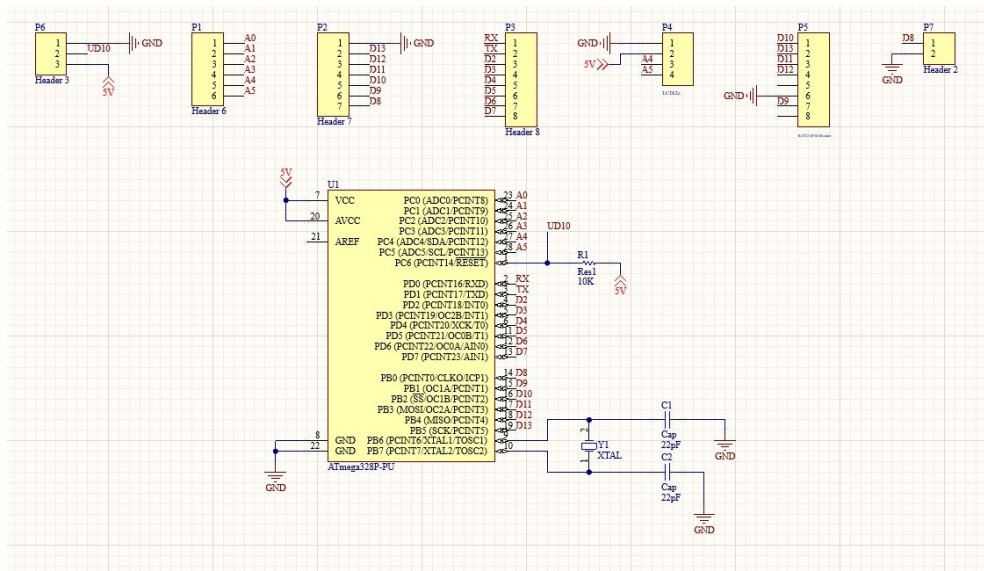


Figure 5.4 RFID Attendance System Using ATMEGA328P (Schematic)

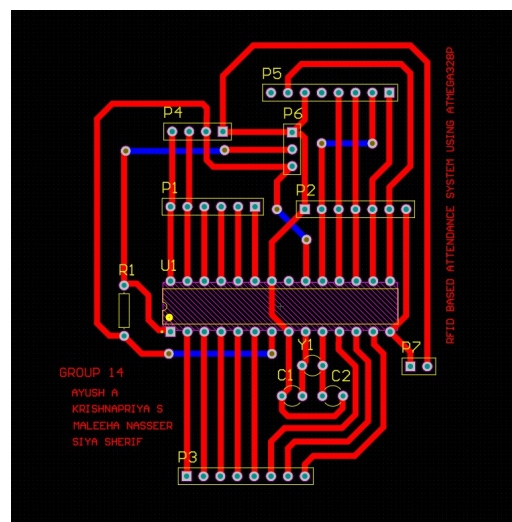


Figure 5.5 RFID attendance system using ATmega328P (2D View)

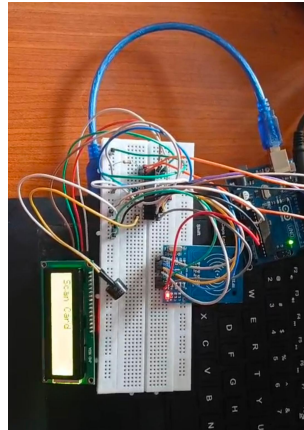
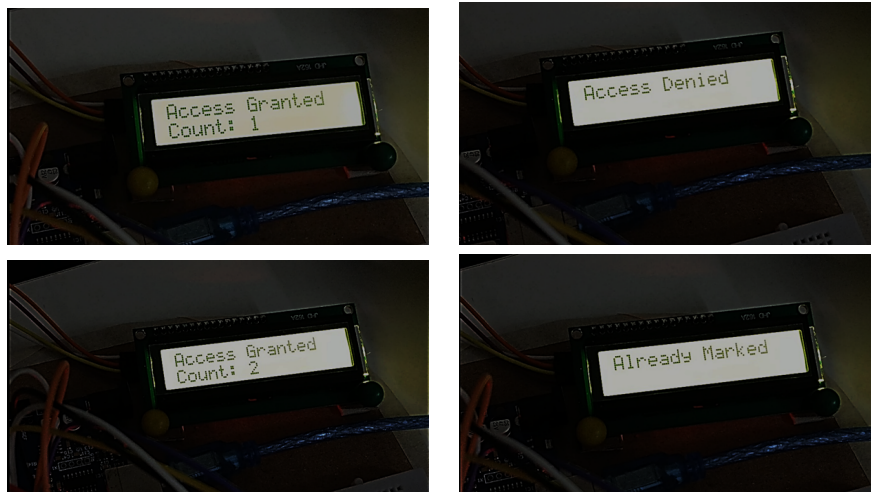


Figure 5.6 Implementation of RFID based attendance system



Figures 5.7 Results obtained from RFID based attendance system

Chapter 6

CONCLUSION AND OVERVIEW

The RFID-based attendance system developed in this project provides a reliable, efficient, and automated solution for attendance management. By utilizing RFID technology with the ATmega328P microcontroller, the system enables accurate and contactless recording of attendance. Each user is identified using a unique RFID tag, and the corresponding data is processed by the microcontroller to mark attendance. The inclusion of an LCD display enhances the usability of the system by providing real-time feedback to the user during operation.

The system effectively reduces the limitations of traditional attendance methods such as manual register entry, which is time-consuming and prone to human errors. The use of RFID technology ensures faster data acquisition and eliminates issues like proxy attendance to a certain extent. The automated nature of the system improves overall efficiency and reliability, making it suitable for use in educational institutions, offices, and other environments where attendance tracking is essential.

Throughout the project, the feasibility and effectiveness of implementing an RFID-based attendance system using a microcontroller have been successfully demonstrated. The integration of hardware components such as the RFID reader, ATmega328P microcontroller, and LCD display illustrates the practical application of embedded systems in real-world scenarios. The system provides a simple yet effective interface for users, ensuring ease of operation and minimal training requirements.

In addition to achieving the intended functionality, the development process offered valuable learning experiences in microcontroller programming, hardware interfacing, and system integration. Various challenges were encountered during the implementation phase, including ensuring proper communication between components and maintaining stable system performance. Addressing these challenges helped in improving problem-solving skills and gaining a deeper understanding of embedded system design.

Although the system meets its primary objectives, there is scope for further improvement and enhancement. Future developments may include the integration of wireless communication modules for remote monitoring, database connectivity for long-term storage of attendance records, and mobile application support for improved accessibility. These enhancements would increase the scalability and functionality of the system, making it more suitable for advanced applications.

REFERENCES

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- SanDisk – SD Card SPI Interface Documentation
- Hitachi – HD44780 LCD Controller Datasheet
- <https://www.hackster.io/ahmedibrrahim/atmega328p-standalone-board-77044d>
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- https://projecthub.arduino.cc/team_chkr/rfid-based-smart-attendance-system-beff99

APPENDIX A

CODE

The following code is used to interface the MFRC522 RFID reader, I2C LCD, and a buzzer with the Arduino UNO to automate the attendance process.

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define SS_PIN 10
#define RST_PIN 9
#define BUZZER 8

MFRC522 mfrc522(SS_PIN, RST_PIN);
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Your UIDs
byte card1[4] = {0x29, 0x8B, 0xFC, 0x03};
byte card2[4] = {0xA3, 0x00, 0xC1, 0x2C};
byte card3[4] = {0xE3, 0xBA, 0xCA, 0x06}; // NEW

bool marked1 = false;
bool marked2 = false;
bool marked3 = false; // NEW

bool checked1 = false;
bool checked2 = false;
bool checked3 = false; // NEW

int count = 0;

void setup() {
  SPI.begin();
  Wire.begin();
  lcd.init();
```

```

lcd.backlight();

pinMode(BUZZER, OUTPUT);

mfrc522.PCD_Init();

lcd.setCursor(0, 0);
lcd.print("Scan Card");
}

void loop() {

    if (!mfrc522.PICC_IsNewCardPresent()) return;
    if (!mfrc522.PICC_ReadCardSerial()) return;

    if (compareUID(mfrc522.uid.uidByte, card1)) {

        if (!marked1) {
            marked1 = true;
            count++;
            accessGranted();
        } else {
            checked1 = true;
            showMessage("Already Marked");
        }

    } else if (compareUID(mfrc522.uid.uidByte, card2)) {

        if (!marked2) {
            marked2 = true;
            count++;
            accessGranted();
        } else {
            checked2 = true;
            showMessage("Already Marked");
        }

    } else if (compareUID(mfrc522.uid.uidByte, card3)) {    // NEW

```

```

    if (!marked3) {
        marked3 = true;
        count++;
        accessGranted();
    } else {
        checked3 = true;
        showMessage("Already Marked");
    }

} else {
    showMessage("Access Denied");
}

delay(2000);

mfrc522.PICC_HaltA();
mfrc522.PCD_StopCrypto1();

// RESET ONLY AFTER ALL 3 RE-SCANNED
if (marked1 && marked2 && marked3 && checked1 && checked2 && checked3) {
    delay(2000);
    resetSystem();
} else {
    lcd.clear();
    lcd.print("Scan Card");
}
}

bool compareUID(byte *a, byte *b) {
    for (byte i = 0; i < 4; i++) {
        if (a[i] != b[i]) return false;
    }
    return true;
}

void accessGranted() {
    lcd.clear();
    lcd.print("Access Granted");
    lcd.setCursor(0, 1);

```

```
    lcd.print("Count: ");
    lcd.print(count);

    tone(BUZZER, 1000);
    delay(200);
    noTone(BUZZER);
}

void showMessage(String msg) {
    lcd.clear();
    lcd.print(msg);
}

void resetSystem() {
    marked1 = false;
    marked2 = false;
    marked3 = false;

    checked1 = false;
    checked2 = false;
    checked3 = false;

    count = 0;

    lcd.clear();
    lcd.print("Scan Card");
}
```

APPENDIX B

DATASHEET

B.1 ATmega328/P

Description

The Atmel AVR[®] core combines a rich instruction set with 32 general purpose working registers. All the 32 registers are directly connected to the Arithmetic Logic Unit (ALU), allowing two independent registers to be accessed in a single instruction executed in one clock cycle. The resulting architecture is more code efficient while achieving throughputs up to ten times faster than conventional CISC microcontrollers.

The ATmega328/P provides the following features: 32 Kbytes of In-System Programmable Flash with Read-While-Write capabilities, 1 Kbyte EEPROM, 2 Kbytes SRAM, 23 general purpose I/O lines, 32 general purpose working registers, Real Time Counter (RTC), three flexible Timer/Counters with compare modes and PWM, 1 serial programmable USART, 1 byte-oriented 2-wire Serial Interface (I2C), a 6 channel 10-bit ADC (8 channels in TQFP and QFN/MLF packages), a programmable Watchdog Timer with internal oscillator, an SPI serial port, and six software selectable power saving modes. The Idle mode stops the CPU while allowing the SRAM, Timer/Counters, SPI port, and interrupt system to continue functioning. The Power-down mode saves the register contents but freezes the oscillator, disabling all other chip functions until the next interrupt or hardware reset. In Power-save mode, the asynchronous timer continues to run, allowing the user to maintain a timer base while the rest of the device is sleeping. The ADC Noise Reduction mode stops the CPU and all I/O modules except asynchronous timer and ADC to minimize switching noise during ADC conversions. In Standby mode, the crystal/resonator oscillator is running while the rest of the device is sleeping. This allows very fast start-up combined with low power consumption. In Extended Standby mode, both the main oscillator and the asynchronous timer continue to run.

Atmel offers the QTouch[®] library for embedding capacitive touch buttons, sliders and wheels functionality into AVR microcontrollers. The patented charge-transfer signal acquisition offers robust sensing and includes fully debounced reporting of touch keys and includes Adjacent Key Suppression[®] (AKS[™]) technology for unambiguous detection of key events. The easy-to-use QTouch Suite toolchain allows you to explore, develop and debug your own touch applications.

The device is manufactured using Atmel's high density non-volatile memory technology. The on-chip ISP Flash allows the program memory to be reprogrammed in-system through an SPI serial interface, by a conventional nonvolatile memory programmer, or by an on-chip boot

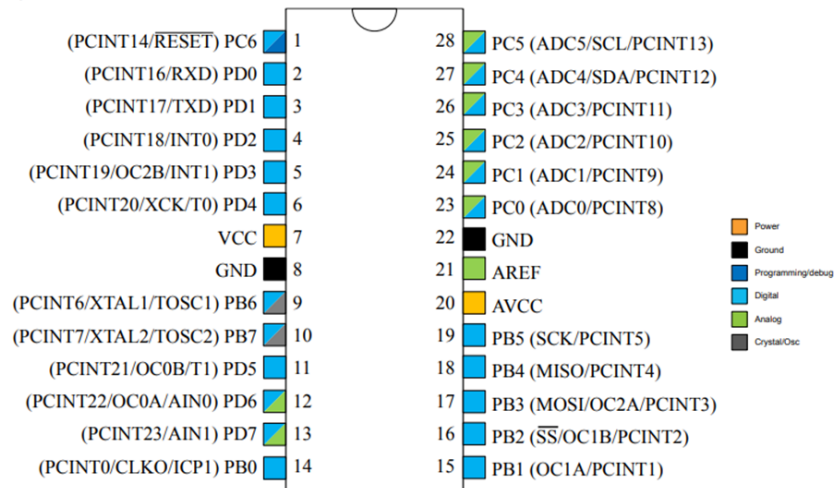


Figure B.3: Pin Configurations

B.4 28-pin MLF Top View

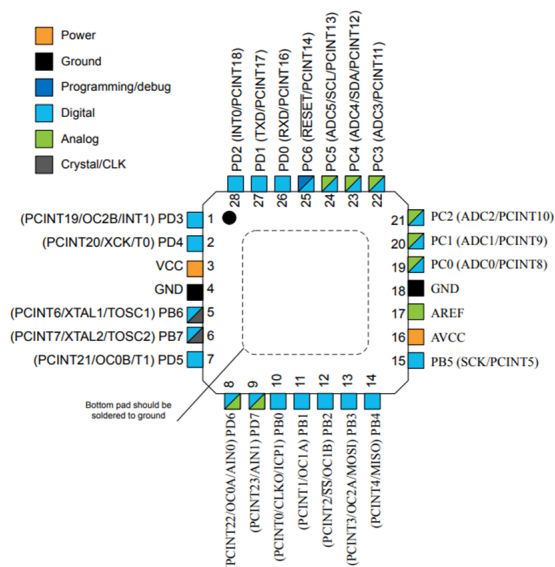


Figure B.4: 28-pin MLF Top View

B.5 32-pin TQFP Top View

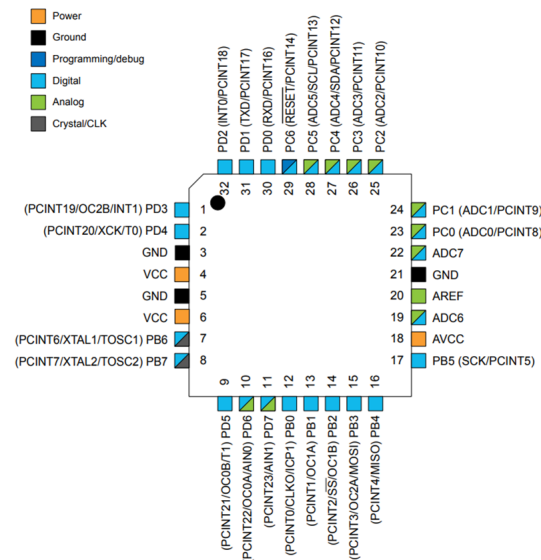


Figure B.5: 32-pin TQFP Top View

B.6 32-pin MLF Top View

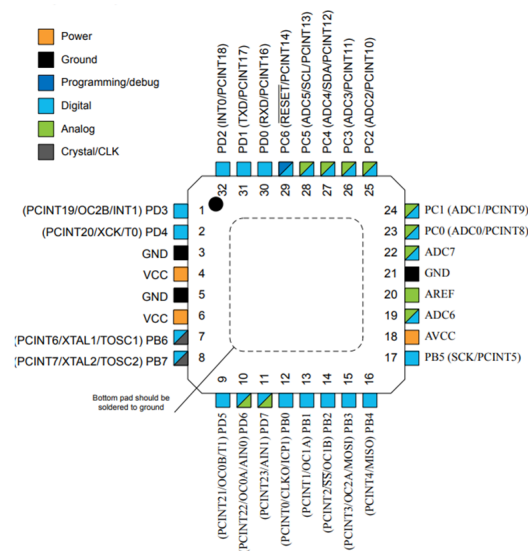


Figure B.6: 32-pin MLF Top View

B.6.1 Pin Descriptions

VCC: Digital supply voltage.

GND: Ground.

PD2 (INT0/PCINT18): External interrupt input pin of Port D.

Port B (PB[7:0]) / XTAL1 / XTAL2 / TOSC1 / TOSC2: Port B is an 8-bit bi-directional I/O port with internal pull-up resistors selected for each bit. The output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, pins externally pulled low will source current when the pull-up resistors are activated. Port B pins are tri-stated when a reset condition becomes active, even if the clock is not running.

Common pins in this group include PD7 (PCINT23/AIN1), PB0 (PCINT0/CLKO/ICP1), PB1 (PCINT1/OC1A), PB2 (PCINT2/SS/OC1B), PB3 (PCINT3/OC2A/MOSI), PB4 (PCINT4/MISO), PB5 (SCK/PCINT5), PC1 (ADC1/PCINT9), PC0 (ADC0/PCINT8), ADC7, ADC6, AREF, and AVCC.

Depending on the clock selection fuse settings, PB6 can be used as input to the inverting oscillator amplifier and to the internal clock operating circuit, while PB7 can be used as output from the inverting oscillator amplifier. If the internal calibrated RC oscillator is used as the chip clock source, PB[7:6] are used as TOSC[2:1] inputs for the asynchronous Timer/Counter2 when the AS2 bit in ASSR is set.

Port C (PC[5:0]): Port C is a 7-bit bi-directional I/O port with internal pull-up resistors selected for each bit. The PC[5:0] output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port C pins that are externally pulled low will source current if the pull-up resistors are activated. Port C pins are tri-stated when a reset condition becomes active, even if the clock is not running.

PC6/RESET: If the RSTDISBL fuse is programmed, PC6 is used as an I/O pin. Note that the electrical characteristics of PC6 differ from those of the other pins of Port C. If the RSTDISBL fuse is unprogrammed, PC6 is used as a reset input. A low level on this pin for longer than the minimum pulse length will generate a reset, even if the clock is not running. Shorter pulses are not guaranteed to generate a reset.

Port D (PD[7:0]): Port D is an 8-bit bi-directional I/O port with internal pull-up resistors selected for each bit. The Port D output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port D pins that are externally pulled low will source current if the pull-up resistors are activated. The Port D pins are tri-stated when a reset condition becomes active, even if the clock is not running.

AVCC: AVCC is the supply voltage pin for the A/D converter, PC[3:0], and PE[3:2]. It should be externally connected to VCC, even if the ADC is not used. If the ADC is used, it should be connected to VCC through a low-pass filter. Note that PC[6:4] use digital supply voltage, VCC.

AREF: AREF is the analog reference pin for the A/D converter.

ADC[7:6] (TQFP and VFQFN package only): In the TQFP and VFQFN package, ADC[7:6] serve as analog inputs to the A/D converter. These pins are powered from the analog supply and serve as 10-bit ADC channels.

B.7 PORT Function Multiplexing**B.7.1 Pins 1–16**

Pin	MLF/TQFP	PDIP	PAD	EXTINT	PCINT	ADC/AC	OSC	T/C #0	T/C #1	USART0	I2C / SPI
1	1	5	PD[3]	INT1	PCINT19			OC2B			
2	2	6	PD[4]		PCINT20			T0	XCK0		
3	3	7	VCC								
4	4	8	GND								
5	-	-	VCC								
6	-	-	GND								
7	5	9	PB[6]		PCINT6		XTAL1/TOSC1				
8	6	10	PB[7]		PCINT7		XTAL2/TOSC2				
9	7	11	PD[5]		PCINT21			OC0B	T1		
10	8	12	PD[6]		PCINT22	AIN0		OC0A			
11	9	13	PD[7]		PCINT23	AIN1					
12	10	14	PB[0]		PCINT0		CLKO	ICP1			
13	11	15	PB[1]		PCINT1			OC1A			
14	12	16	PB[2]		PCINT2			OC1B			SS0
15	13	17	PB[3]		PCINT3			OC2A			MOSI
16	14	18	PB[4]		PCINT4						MISO

B.7.2 Pins 17–32

Pin	MLF/TQFP	PDIP	PAD	EXTINT	PCINT	ADC/AC	OSC	T/C #0	T/C #1	USART0	I2C / SPI
17	15	19	PB[5]		PCINT5						SCK0
18	16	20	AVCC								
19	-	-	ADC6			ADC6					
20	17	21	AREF								
21	18	22	GND								
22	-	-	ADC7			ADC7					
23	19	23	PC[0]		PCINT8	ADC0					
24	20	24	PC[1]		PCINT9	ADC1					
25	21	25	PC[2]		PCINT10	ADC2					
26	22	26	PC[3]		PCINT11	ADC3					
27	23	27	PC[4]		PCINT12	ADC4					SDA0
28	24	28	PC[5]		PCINT13	ADC5					SCL0
29	25	1	PC[6]/RESET		PCINT14						
30	26	2	PD[0]		PCINT16					RXD0	
31	27	3	PD[1]		PCINT17					TXD0	
32	28	4	PD[2]	INT0	PCINT18						

B.8 RFID RC522

Mifare RC522 is a highly integrated RFID card reader that works on non-contact 13.56 MHz communication. It is designed by NXP as a low-power, low-cost, and compact read/write chip, making it a suitable choice for the development of smart meters and portable hand-held devices.

MFRC522 uses an advanced modulation system and is fully integrated at 13.56 MHz with various passive non-contact communication protocols. It supports 14443A compatible answer signals. The DSP handles ISO14443A frames and error correction. Furthermore, it supports rapid CRYPTO1 encryption to validate Mifare series products.

MFRC522 supports higher-speed non-contact communication for Mifare series products, with duplex communication speeds up to 424 kb/s. As a member of the 13.56 MHz RFID family, MFRC522 has many similarities to MF RC5200 and MFRC530, while also offering additional new features.

This module can fit directly into hand-held devices for mass production. The module uses a 3.3 V power supply and can communicate directly with any CPU board through the SPI protocol, ensuring reliable operation and a good reading distance.

RC522 Pin Configuration:

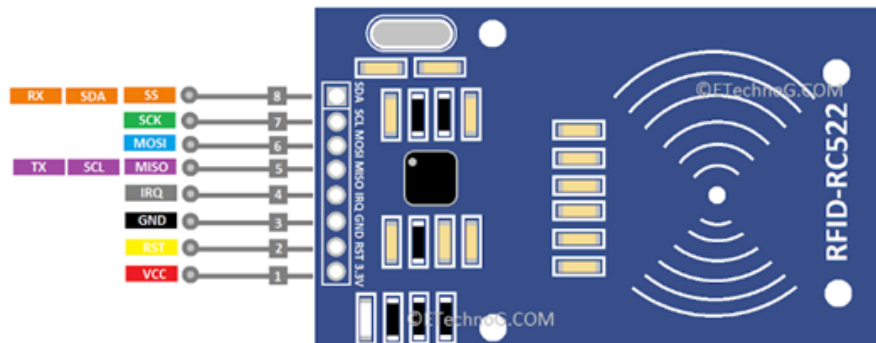


Figure B.8: RFID

SPECIFICATIONS

Operating Frequency: 13.56 MHz

Operating Voltage: 2.5 V to 3.3 V

Communication: SPI, I²C protocol, UART

Maximum Data Rate: 10 Mbps

Read Range: 5 cm

Current Consumption: 13–26 mA

Power Down Mode Consumption: 10 μ A (min)

Sleep Current: < 80 μ A

Peak Current: < 30 mA

Supported Card Types: MIFARE S50, MIFARE S70, MIFARE Ultralight, MIFARE Pro, MIFARE DESFire

Dimensions: 40 mm × 60 mm

Module Interface SPI Data Transfer Rate: Max. 10 Mbit/s **RC522 Pin Configuration:**

Pin Number	Pin Name	Description
1	VCC	Used to power the module; typically 3.3 V is used.
2	RST	Reset pin – used to reset or power down the module.
3	GND	Connected to ground of the system.
4	IRQ	Interrupt pin – used to wake up the module when a device comes into range.
5	MISO / SCL / TX	MISO for SPI, SCL for I ² C, TX for UART.
6	MOSI	Master Out Slave In pin for SPI communication.
7	SCK	Serial Clock pin – provides clock source.
8	SS / SDA / RX	SS for SPI, SDA for I ² C, RX for UART.

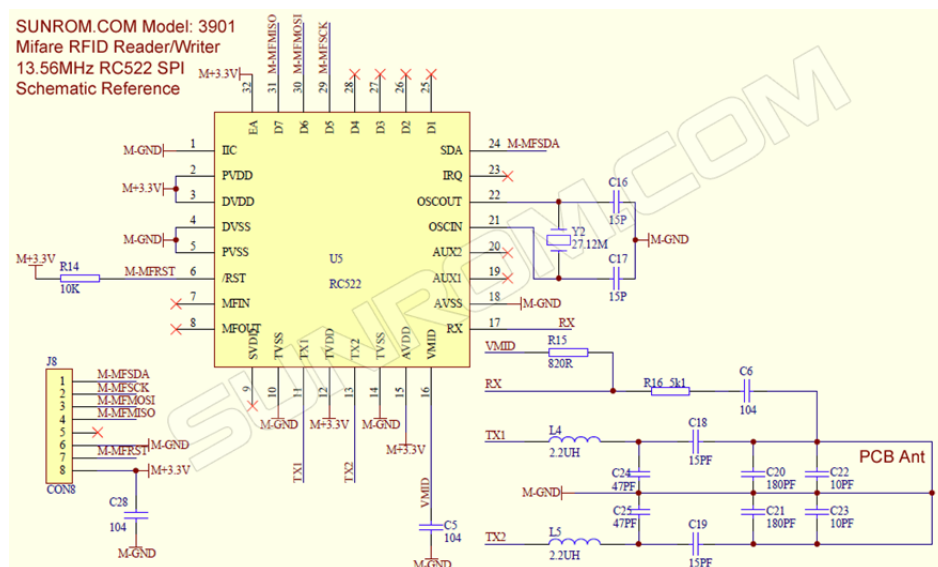


Figure B.9: Schematic of RFID RC522

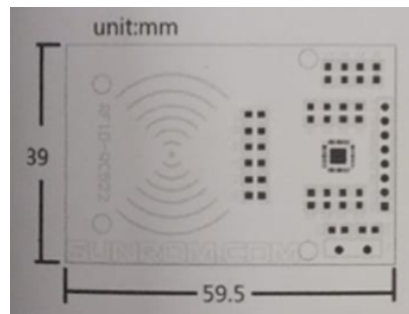


Figure B.10: RFID PCB Schematic